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Master in Mathematics

DFTK Refinement: Verification Across a Large Set of Systems

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Abstract

We implement the error estimates from Cancès et al. [5] into DFTK and evaluate them across a broad set of insulating systems. By comparing Birch–Murnaghan equation-of-state metrics computed with a reduced cutoff (E_{cut}) plus its error estimate against a high-cutoff reference, we quantify the errors and derive guidelines for choosing parameters.

1 Introduction

Density-functional theory (DFT) is a widely used computational approach for modeling the quantum behavior of electrons in materials. It offers a first-principles calculation of optical, mechanical, and chemical properties while keeping computational costs manageable. For most systems, DFT delivers reliable predictions, and its limitations are well understood [13, 3, 4].

Its predictive accuracy hinges critically on the completeness of the underlying basis set. In plane-wave approaches the basis-set quality is controlled by the kinetic-energy cutoff E_{cut} (see section 2.1 for notation), and higher cutoffs improve convergence but increase the computational cost roughly as $E_{\text{cut}}^{3/2}$ for a fixed k-point mesh[16, 13]. In 2021, Cancès et al. introduced a practical scheme for plane-wave DFT that provides computable error bounds on ground-state densities, energies, and forces [5]. The method performs the bulk of the calculation at a modest cutoff and then applies a one-Newton-step post-processfing on a finer grid. Throughout this report, we refer to this last step as *refinement*. Although the linearization approximation is excellent for silicon, gallium arsenide and titanium dioxide [5], its reliability across chemically and structurally diverse materials has not been quantified systematically.

The Density-Functional Toolkit (DFTK) is a Julia library for plane-wave DFT [10]. Bruno Ploumhans recently implemented the refinement protocol for insulating systems. This work addresses the following question:

> *How well does DFTK's refinement protocol reproduce high-cutoff Birch–Murnaghan equation-of-state metrics for insulating crystals, and how do the residual errors scale with the energy gap between low- and high-cutoff calculations?*

Equations of state are a standard tool for studying convergence in DFT. Following Lejaeghere et al.[12] and Bosoni et al. [2], we use the Birch–Murnaghan equation of state (EOS) to quantify the accuracy of the refinement protocol using specific metrics Δ, ε and ν (see section 2.4 for details).

1.1 Scope and limitations

We analyze how Birch–Murnaghan EOS parameters and total energies converge with respect to the initial, lower–cutoff stage of the refinement protocol. The refinement stage itself uses a fixed, higher cutoff to ensure final accuracy. We test on the same set of insulators oxide materials used by Bosoni et al.[2] (see section 3.1 for precisions), using the PBE exchange–correlation functional and PseudoDojo pseudopotentials [21](version 0.5, standard UPF).

Our focus is on nonmagnetic insulating systems at zero temperature with static EOS calculations—atomic positions and cell shapes are not reoptimized at each volume when studying E_{cut} convergence. Note that the refinement protocol of Cancès et al. [5] is formulated specifically for insulators at zero temperature and does not account for errors due to finite Brillouin-zone (k-point) sampling. Although the workflow can handle relaxation and spin polarization, we do not explore their interaction with this E_{cut} refinement protocol here. We also do not include Hubbard U corrections or explicit van der Waals functionals in this analysis. Consequently, our conclusions about cutoff convergence apply directly to DFTK under these conditions (and with these pseudopotentials) and may not hold for other codes, pseudopotential types, or more complex scenarios (e.g., strong correlation or fully relaxed magnetic systems) without further study.

2 Notation and Definitions

2.1 Discretization

For every vector G of the reciprocal lattice $\mathcal{R}^*$ we introduce the Fourier mode

$$e_G(x) = \frac{1}{\sqrt{|\Gamma|}} \exp(\mathrm{i}G \cdot x) \qquad \forall x \in \mathbb{R}^3$$

where Γ is the real-space unit cell and $|\Gamma|$ its Lebesgue measure. The family $\{e_G\}_{G\in\mathcal{R}^*}$ forms an orthonormal basis of $L^2_\#$, the space of square-integrable $\mathcal{R}$-periodic functions.

In plane-wave discretisations of Kohn–Sham DFT the Hilbert space is truncated to the finite sub-space

$$X_{E_{\text{cut}}} = \mathrm{Span}\Big\{e_G : G \in \mathcal{R}^*, \tfrac{1}{2}|G|^2 \le E_{\text{cut}}\Big\}$$

where $E_{\text{cut}} > 0$ is the kinetic-energy cut-off that defines the resolution of the calculation. Raising E_{cut} enlarges the variational space, systematically reducing the discretization error while introducing higher-frequency plane waves that resolve the rapid oscillations near atomic cores.

2.2 Mean-Field Model

After setting a cutoff energy, the truncated finite dimensional space $X_{E_{\text{cut}}}$ is identified[1] with $\mathbb{C}^N$, where N is given by:

$$N = \dim(X_{E_{\text{cut}}}) = \#\left\{G \in \mathcal{R}^*, \tfrac{1}{2}|G|^2 \leq E_{\text{cut}}\right\}$$

Let $\mathcal{H} \subset \mathbb{C}^{N\times N}$ denote the real vector space of Hermitian matrices, with Frobenius inner product. The mean-field model in density-matrix form is the subspace optimization problem:

$$\min_{P\in\mathcal{M}_N} E(P) \tag{1}$$

where $\mathcal{M}_N := \{P \in \mathcal{H} \mid P^2 = P, \mathrm{Tr}(P) = N_{\text{el}}\}$ with N_{el} the fixed number of electrons (or electron pairs) and $E : \mathcal{H} \to \mathbb{R}$ is a C^2 nonlinear energy functional (For more details about the connection between this formulation and Kohn–Sham theory, see [5]).

Solving (1) has been a core task since the beginnings of quantum mechanics. The earliest—and still standard—algorithm is the self-consistent field (SCF) scheme [18, 14]. We'll work with its implementation within DFTK [10]. In practice, solving (1) via (SCF) iterations yields a ground-state projector P_*, which is assumed to be unique in the approach introduced by Cancès et al [5] (for convergence analysis of the SCF iterations see [6]).

2.3 Refinement

The objective of the errors approximation from Cancès et al [5] is—given $P \in \mathcal{M}_N$ close enough to the ground state projector P_*—to approximate the first order correction (*refinement*) of $\delta P = P - P_*$. Considering P in the vicinity of P_*, we have [5]:

$$P - P_* = \Pi_P(P - P_*) + O(\|P - P_*\|_F^2) \tag{2}$$

where Π_P is the orthogonal projection onto $T_P\mathcal{M}_N$. The residual of the problem defines a smooth vector field on $\mathcal{M}_N$ (being zero at P_*):

$$R(P) = \Pi_P H(P) \in T_P\mathcal{M}_N$$

where H is the mean-field Hamiltonian. Cancès et al [6] introduce two superoperators $\mathbf{\Omega}, \mathbf{K}$ such that $\mathbf{\Omega}(P_*) + \mathbf{K}(P_*)$ is the Jacobian of the map $P \mapsto R(P)$ at P_*. Therefore the optimality condition $R(P_*) = 0$ and the expansion (2) yield:

$$R(P) = (\mathbf{\Omega}(P) + \mathbf{K}(P))\Pi_P(P - P_*) + O(\|P - P_*\|_F^2)$$

Using again (2) and the fact that $R(P) \in T_P\mathcal{M}_N$, one can obtain after inversion of $\mathbf{\Omega}(P) + \mathbf{K}(P)$ on the tangent space [5]:

$$\begin{aligned} P - P_* = &\left((\Omega(P) + K(P))|_{T_P\mathcal{M}_N}\right)^{-1} R(P) \\ &+ O(\|P - P_*\|_F^2) \end{aligned} \tag{3}$$

[1] we identify $X_{E_{\text{cut}}}$ with $\mathbb{C}^N$ by labelling the reciprocal-lattice vectors from 1 to N in such a way that, for all $1 \leq i < j \leq N$, $|G_i| \leq |G_j|$;

Base on this expansion it is possible to formulate a Newton algorithm to solve $R(P_*) = 0$:

$$P^{k+1} = \mathcal{R}_{P^k}\Big(P^k - (\Omega^k + K^k)^{-1}R(P^k)\Big) \tag{4}$$

where $\Omega^k := \Omega(P^k)\big|_{T_{P^k}\mathcal{M}_N}$, $K^k := K(P^k)\big|_{T_{P^k}\mathcal{M}_N}$, and $\mathcal{R}$ is a suitable retraction on $\mathcal{M}_N$ (see [6] for such possible retraction). The cost of a Newton step is comparable to that of a full self-consistent field cycle [5]. Cancès et al proposed a way to improve the error estimate by decomposing tangent vectors and operators according to the splitting $\Pi_{E_{\text{cut}}}T_P\mathcal{M}_N$ and $\Pi^{\perp}_{E_{\text{cut}}}T_P\mathcal{M}_N$, which will be denoted as 1 and 2 (section 5.2. of [5]). The error residual relation (3) can thus be written as:

$$\begin{pmatrix} (\mathbf{\Omega} + \mathbf{K})_{11} & (\mathbf{\Omega} + \mathbf{K})_{12} \\ (\mathbf{\Omega} + \mathbf{K})_{21} & (\mathbf{\Omega} + \mathbf{K})_{22} \end{pmatrix} \begin{pmatrix} P_1 - P_{*1} \\ P_2 - P_{*2} \end{pmatrix} = \begin{pmatrix} R_1 \\ R_2 \end{pmatrix}$$

In practice inverting $\mathbf{\Omega} + \mathbf{K}$ would cost as much as a full Newton step on the reference grid. But one can make approximations in order to invert it only on the coarser grid $X_{E_{\text{cut}}}$. Following the perturbation-theory-based postprocessing method [7, 9] and the Feshbach-Schur method [8], one can approximate:

$$(\mathbf{\Omega} + \mathbf{K})_{21} \approx 0 \quad (\mathbf{\Omega} + \mathbf{K})_{22} \approx \mathbf{M}_{22}$$

see [5] for a precise definition of the operator $\mathbf{M}$, note that one can compute its inverse effectively. This finally gives us the system:

$$\begin{aligned} P_2 - P_{*2} &\approx \mathbf{M}_{22}^{-1}R_2 \\ P_1 - P_{*1} &\approx (\mathbf{\Omega} + \mathbf{K})_{11}^{-1}(R_1 - (\mathbf{\Omega} + \mathbf{K})_{12}\mathbf{M}_{22}^{-1}R_2) \end{aligned} \tag{5}$$

which requires only a single efficient computation on the finer grid. The DFTK implementation of *refinement* follows this two-space splitting [10] method.

2.4 Birch-Murnaghan EOS

For crystalline solids, the response to a hydrostatic compression can be described by an Eulerian finite strain

$$f = \tfrac{1}{2}\Big[(V/V_0)^{-2/3} - 1\Big]$$

where V is the instantaneous volume and V_0 the equilibrium volume. Expanding the total energy $E(V)$ in powers of f and truncating at third order yields the *Birch–Murnaghan equation of state* (EOS) [1, 15]:

$$E(V) = E_0 + \frac{9V_0B_0}{16}\left\{\Big[f^3(B_0' - 4)\Big] + \Big[f^2(6 - 4f)\Big]\right\} \tag{6}$$

Here E_0 is the minimum energy, B_0 the bulk modulus at V_0, and $B_0' = (\mathrm{d}B/\mathrm{d}P)_0$ its first pressure derivative. We follow the same approach as Lejaeghere et al.[12] and Bosoni et al.[2] by performing a fit of $E(V)$ of equation (6); using calculations of the total energy corresponding to 7 equidistant constant volumes between 94% and 106% for each structure of a reference

central volume $\tilde{V}_0$. It is thus possible to extract theoretical prediction of V_0, B_0 and B_0'.

To quantify the difference between two fitted EOS curves, $E_a(V)$ and $E_b(V)$, we introduce three scalar metrics Δ (introduced in [12]), ε and ν (introduced in [2]). Firstly, ε represents the area between the two EOS curves normalized by the average variance of the two curves. Over the volume interval $[V_m, V_M]$, i.e. $V_m = 94\%\tilde{V}_0$ and $V_M = 106\%\tilde{V}_0$, let's define

$$\langle f \rangle = \frac{1}{V_M - V_m} \int_{V_m}^{V_M} f(V)\,\mathrm{d}V$$

Then

$$\varepsilon(a,b) = \sqrt{\frac{\langle [E_a(V) - E_b(V)]^2 \rangle}{\sqrt{\langle [E_a(V) - \langle E_a \rangle]^2 \rangle \langle [E_b(V) - \langle E_b \rangle]^2 \rangle}}}$$

An excellent agreement corresponds to $\varepsilon < 0.06$ and a good agreement to $\varepsilon < 0.20$ [2].

Then ν captures the relative differences in equilibrium volume V_0, bulk modulus B_0, and pressure derivative B_0'. Denote by $(V_0)_a$, $(B_0)_a$, $(B_0')_a$ the values obtained from fitting $E_a(V)$, and similarly for method b, define

$$\nu(a,b) = 100 \sqrt{\sum_{Y \in \{V_0, B_0, B_0'\}} \left[w_Y \frac{Y_a - Y_b}{(Y_a + Y_b)/2} \right]^2}$$

with weights $w_{V_0} = 1$, $w_{B_0} = 0.0568$, $w_{B_0'} = 0.2755$. An excellent agreement satisfies $\nu < 0.04$ and a good agreement $\nu < 0.15$, see appendix A for a discussion about the choice of those weights and treshold.

Finally Δ represents the area between the two EOS curves:

$$\Delta(a,b) = \sqrt{\langle [E_a(V) - E_b(V)]^2 \rangle}$$

This metric has the downside of being too sensitive to the value of the bulk modulus of the material [2]. The values are normalized by the number of atoms. When the bulk modulus is not too small, an excellent agreement is $\Delta < 0.30\,\mathrm{meV/atom}$, while a good agreement is $\Delta < 0.95\,\mathrm{meV/atom}$ [2].

3 Computational Setup

All calculations presented in this work were managed and automated using the AiiDA framework [17], an open-source Python infrastructure designed to aid researchers in managing, preserving, and disseminating computational workflows. The specific workflows and protocols for material property calculations were primarily sourced from the `aiida-common-workflows` (ACWF) package which has beeen adapted to work with the *refinement* protocol by Bruno Ploumhans [11, 19]. Structure generation, job submission, and post-processing are all carried out with a customised fork of `acwf-verification-scripts`, which has been further extended to support the *refinement* workflow [2, 19].

The core of this study involves the calculation of the Birch-Murnaghan EOS. This was achieved by employing the `EquationOfStateWorkChain` provided by ACWF. This high-level workflow orchestrates a series of calculations where the volume of the input crystal structure is systematically varied, and for each volume point, the total energy is computed, Figure 1 showcases a simplified version of the workflow.

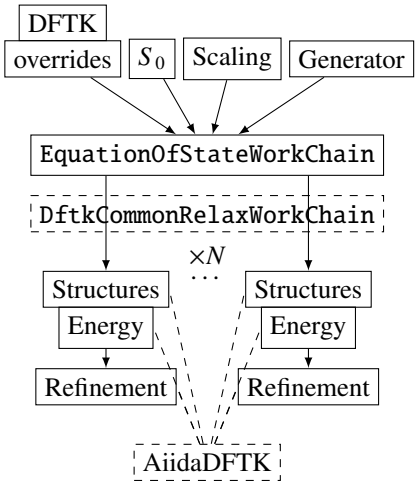

Figure 1: Simplified visualization of the workflow. The `EquationOfStateWorkChain` takes several arguments: "DFTK" is the plugin used, with "overrides" to change certain calculation parameters (typically the E_{cut}); "S_0" denotes the equilibrium structures; "Scaling" defines the volume scaling (here from $94\%V_0$ to $106\%V_0$ in $N = 7$ steps); and "Generator" specifies items such as the protocol (where the refinement is chosen), the electronic type, and the engine settings. The `EquationOfStateWorkChain` then submits calculations for each volume scaling, and relaxation is performed by `DftkCommonRelaxWorkChain` to obtain new structures (in our case, no relaxation is applied). Total energies are then computed, and the refinement step is carried out on a finer grid.

For DFT calculations within each step of the equation of state workflow, the DFTK code was utilized. The interface between AiiDA and DFTK is facilitated by the `aiida-dftk` plugin [20]. Specifically, the `EquationOfStateWorkChain` delegates the individual energy/relaxation calculations to the DFTK-specific implementation of the common relaxation workflow, namely the `DftkCommonRelaxWorkChain`.

The central interest is the *refinement* protocol available for DFTK within ACWF. This protocol involves a two-stage computational approach for each volume point in the EOS:

1. An initial self-consistent field (SCF) calculation is performed with a set of parameters corresponding to a lower, computationally cheaper energy cutoff for the plane-wave basis set (hereafter referred to as E_{cut}).

2. A subsequent refinement step is then performed with significantly higher, fixed energy cutoff (hereafter $E_{\text{cut}}^{\text{high}}$).

The `DftkCommonRelaxInputGenerator` is responsible for preparing the inputs for these stages based on the selected *refinement* protocol.We set $E_{\text{cut}}^{\text{high}}$ to twice the pseudopotential family's "high" recommendation, ensuring valid results at the end of the E_{cut} sweep.

We then used a script launching a series of `EquationOfStateWorkChain` calculations. For each launch, the E_{cut} used in the initial stage of the DFTK *refinement* protocol is varied from $0.025 \cdot E_{\text{cut}^{\text{high}}}$ to $0.5 \cdot E_{\text{cut}}^{\text{high}}$. We sampled E_{cut}-factor values, $\{E_{\text{cut}}\text{-factor}\}_{i=1}^{N} \subset [0.025, 0.5]$, which multiply the high cutoff $E_{\text{cut}}^{\text{high}}$. This lets us compare convergence plots across different structures. Plotting convergence directly against E_{cut} can be misleading if each structure needs a different cutoff to start converging. By normalizing with these factors, we compare all structures on the same relative scale.

The $E_{\text{cut}}^{\text{high}}$ itself remains constant across all these scans. This allows for an examination of how the final EOS converge as the initial E_{cut} approaches the $E_{\text{cut}}^{\text{high}}$. For comparison, a baseline set of calculations using a fixed $E_{\text{cut}}^{\text{high}}$ was also performed using the same workflow as described in Figure 1 but without the refinement step.

All DFTK calculations were performed on the Helvetios HPC cluster at EPFL. AiiDA managed the submission of jobs to the Slurm workload manager, with computational resources (number of MPI processes, walltime limits) specified within the launch scripts (e.g., `NUM_PROCS = 6` and `max_wallclock_seconds = 3600`).

3.1 Benchmark set

The present study focuses on a subset of *37 binary oxide prototypes*, see Table 1. Because the refinement protocol currently applies only to insulators [5], we selected insulating structures from the verification dataset [2]. Although some of these systems yielded an insufficient number of converged volume points for a robust Birch–Murnaghan fit at certain E_{cut} values (e.g., SeO_2 and TeO_2 failed to converge across multiple E_{cut}). The majority still yielded enough points for a reliable analysis. Any resulting gaps in the statistical analysis are explicitly visible (for example, the plot will be explicitly disconnected at those points).

Stoichiometry pattern	
XO:	BaO, BeO, CO, CaO, GeO, KO, MgO, PbO, SiO, SnO, SrO, ZnO
X_2O:	Be_2O, Ca_2O, Cs_2O, H_2O, Li_2O, Mg_2O, Na_2O, Rb_2O, Sr_2O, Tl_2O, Zn_2O
XO_2:	FeO_2, HfO_2, OsO_2, PoO_2, RuO_2, SeO_2, SiO_2, TeO_2, TiO_2, ZrO_2
X_2O_3:	Ir_2O_3
XO_3:	RnO_3, WO_3

Table 1: List of the 37 insulating structures included in this report's benchmark.

4 Results and Discussion

4.1 Parameters Convergence

The first step in validating the protocol refinement results is to check the convergence of the equation-of-state fit parameters (Eq. 6), namely E_0, V_0, B_0, and B_0'. Although this approach is not the most rigorous validation, nonetheless, these parameters are useful indicators. We will address the use of EOS metrics in section 4.2.

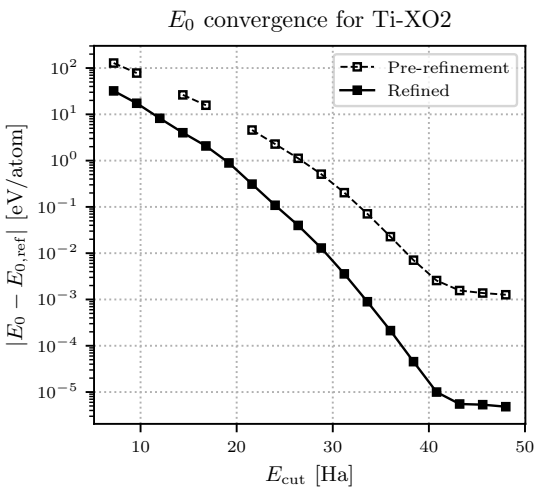

Figure 2: Absolute difference of E_0 from the reference value computed at a high cutoff $E_{\text{cut}}^{\text{high}}$, plotted as a function of E_{cut}. Energies are scaled with respect to the number of atoms in the simulation cell. The refinement method is shown with a solid line; the SCF-only method with a dashed line. The refinement method demonstrates clear convergence advantages across the tested E_{cut} range.

Figure 2 shows the convergence of E_0 for the TiO_2 structure as a function of the cutoff energy E_{cut}. The solid line corresponds to the refined pro-

tocol, while the dashed line represents the standard self-consistent field method without refinement. As shown, the refinement method converges faster and levels off at a higher precision. This behavior seems consistent with the trend reported by Cancès et al. (Section 6, Ref. [5]), who also studied TiO_2. However, note that their results use the Newton method (Eq. 4), whereas our implementation uses the refinement method (Eq. 5).

As noted above, there are gaps in the plot where the EOS fit failed for the SCF-only method. The refinement protocol fills these gaps, demonstrating for the first time that even when the SCF-only method cannot produce representative results, refinement can partially recover them. This will be discussed a bit further in Section 4.2.

However, the plots for the other parameters of the EOS are less smooth. Figures 16 and 17 in Appendix B show the convergence of B_0 and B_0' for the TiO_2 structure, respectively, as a function of E_{cut}. The refinement method also demonstrates clear convergence advantages across the tested E_{cut} range.

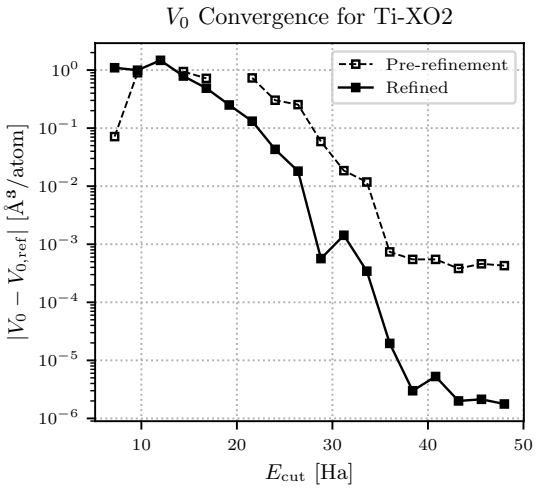

Figure 3: Absolute difference of V_0 from the reference value computed at a high cutoff $E_{\text{cut}}^{\text{high}}$, plotted as a function of E_{cut}. Volumes are scaled with respect to the number of atoms in the simulation cell. The refinement method is shown with a solid line; the SCF-only method with a dashed line. The refinement method demonstrates clear convergence advantages across the tested E_{cut} range.

Figure 3 shows the convergence of V_0 for the TiO_2 structure as a function of E_{cut}. Again, the refinement method converges faster than the SCF-only method. For the lowest E_{cut}, the SCF-only method yields a slightly better value, but at such a low cutoff the errors in other parameters are large: the E_0 error is on the order of 10^2 (Fig. 2), the B_0 error is of order 1 (Fig. 16), and the B_0' error is on the order of 10^2 (Fig. 17), making this initial improvement insignificant. Similarly, although the SCF-only method yields a slightly better B_0 at the lowest E_{cut} in Fig. 16, the errors remain too large to be significant.

In Fig. 3, at the highest E_{cut} values, the refinement protocol shows a "bounce back," with its precision suddenly worsening. This phenomenon was observed across multiple structures, where under certain conditions the refinement protocol loses the extra precision it provided. It is most apparent in Figure 4 for the Ti_2O structure, where the refinement bounces back, levels off, and is surpassed in precision by the SCF-only method.

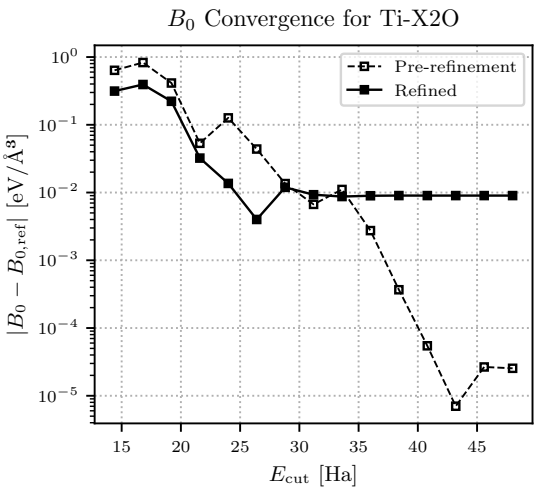

Figure 4: Absolute difference of B_0 from the reference value computed at a high cutoff $E_{\text{cut}}^{\text{high}}$, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. After a certain cutoff, the SCF-only method outperforms refinement in precision.

Figure 18 and 19 showcasing this happening for B_0 for the structure CO and B_0' for the structure Ca_2O showing this happening for different structures and parameters. Importantly, whenever this behavior appears, it occurs consistently for V_0, B_0, and B_0' (see Fig. 20 for B_0 and Fig. 21 for B_0' in the appendix for the Ti_2O structure).However, this never occurs for the energies E_0 (Fig. 22 for the Ti_2O structure), suggesting that while the refinement method can approximate the energy accurately, its fitting over volume changes may be inconsistent.

This suggests that, for certain structures, the approximation made to obtain Eq. 5 may break down. The structures for which the refinement exhibit this strange behaviour are the following: Be_2O, CO, FeO_2, RnO_3, Ti_2O and Zn_2O. In Section 4.3, we will explore whether we can identify a pattern where refinement works well, but for now we'll stay focused on the large-scale convergence of the parameters.

We generated box plots showing the median, interquartile range (IQR), and outliers. Figure 5

presents the convergence of the EOS parameters with respect to the $\{E_{\text{cut}}\text{-factor}\}_{i=1}^{N}$. The plots show the distribution of E_0,V_0,B_0 and B_0' across all computed materials, relative to a well-converged high-cutoff reference. For each parameter, we compare the results obtained from the standard low-cutoff DFT calculation ("Pre-refinement", grey) against those obtained after applying our energy correction scheme ("Refined", black).

A clear and consistent trend emerges from all four panels. The error distributions for the refined results are systematically lower than their pre-refinement counterparts at every E_{cut}-factor. This demonstrates that the refinement procedure substantially improves the accuracy of the computed EOS parameters. More significantly, the refinement scheme drastically accelerates the convergence. For instance, the median of the refined V_0 B_0 and B_0' at a E_{cut}-factor = 0.225 is comparable to, or even better than, the median error of the pre-refinement calculations at a factor of 0.325 or higher. This indicates that our method allows us to achieve high-precision results.

In summary, these results provide strong evidence that the post-processing refinement is effective at reducing errors. Overall, the refinement protocol produces more consistent energy results: the convergence curve is smoother, the interquartile range remains narrow, and outliers stay close to the central spread. In contrast, for V_0, B_0, and B_0', refinement's interquartile range grows noticeably with increasing E_{cut}-factor, while the SCF-only IQR stays relatively constant. This effect is most striking for B_0', where refinement's IQR spans over two orders of magnitude at the highest E_{cut}-factors, compared to less than one order for SCF-only. Outliers for these three parameters also lie much farther from the median—often by a factor of three—whereas energy outliers remain close to the box edges. Although SCF-only shows some extreme values (for example, a low outlier at E_{cut}-factor = 0.35 for B_0), these are less pronounced than in the refinement distributions.

4.2 Metric Convergence

While analyzing the convergence of individual EOS parameters provides detailed insight, it is often more practical to assess the overall agreement between two equations of state using a single, comprehensive metric. To this end, we now examine the convergence of the Δ, ε and ν metrics, previously introduced in Section 2.4. Figure 6 shows the convergence of ν versus E_{cut} for Li_2O. As expected, the refinement curve converges faster than the SCF-only curve. The "good" threshold is reached at $E_{\text{cut}} \geq 16.8$ Ha, and the "excellent" threshold at $E_{\text{cut}} \geq 22$ Ha. It's surprising that this metric reports "good" so early, especially compared to the convergence of E_0, V_0, B_0, and B_0', which take some time to have an order below 1.

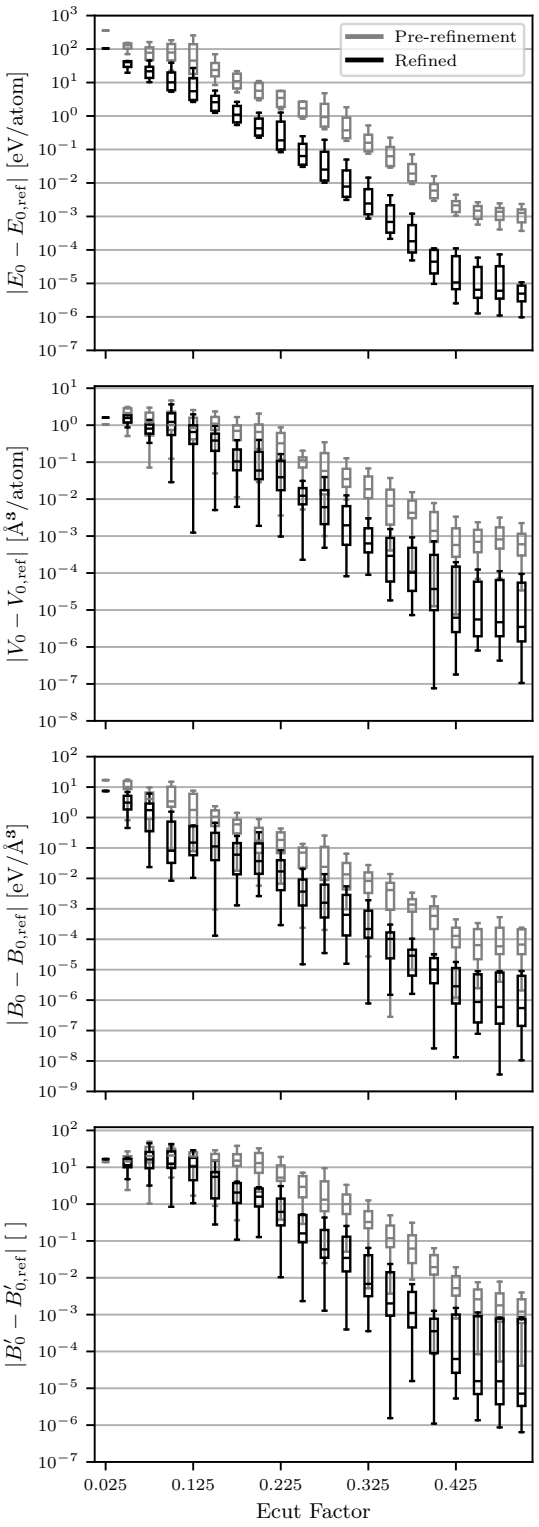

Figure 5: Box plot of E_0, V_0, B_0 and B_0' deviations versus cutoff factor $\{E_{\text{cut}}\text{-factor}_i\}_{i=1}^{N}$, showing the median, interquartile range, and outliers for all structures to compare parameters convergence on a common relative scale. The refinement method is shown in black; the SCF-only method in gray.

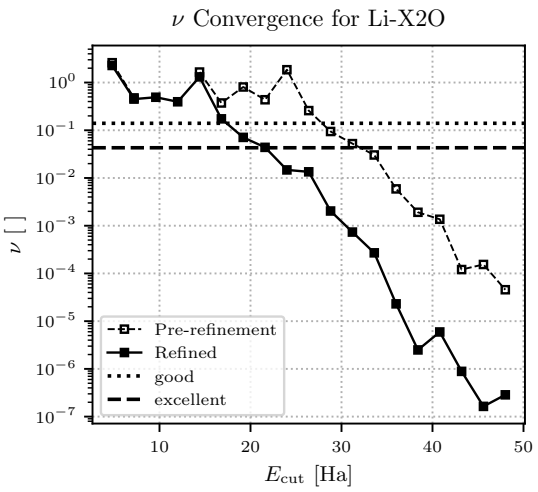

Figure 6: Convergence of the metric ν, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. The refinement method demonstrates clear convergence advantages across the tested E_{cut} range.

One way to check whether the "good" designation is misleading is to plot the EOS curves at a critical cutoff where the threshold is reached (or almost). Figure 7 shows the reference EOS, the refined EOS, and the unrefined EOS at E_{cut} = 16.8 Ha.

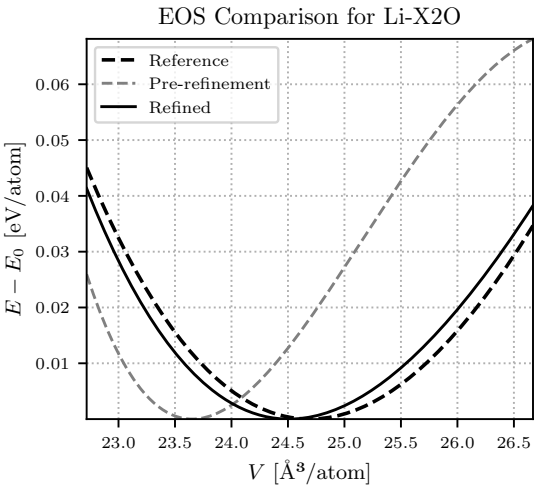

Figure 7: Birch-Murnaghan fit for the reference $E_{\text{cut}}^{\text{high}}$ for the refinement protocol and for the unrefined process at E_{cut} = 16.8 Ha.

As shown, the refined EOS curve isn't perfect, but it brings the unrefined curve much closer to the reference. However, Figures 23 and 24 in the appendix show that, at this cutoff, the curve still doesn't meet the "good" standard under the ε and Δ metrics. This highlights a key issue: the ν metric is less strict and flags curves as "good" more easily. I haven't observed a clear misclassification by ν, and Fig. 7 confirms that even at critical E_{cut}, the refined EOS remains reasonably close to the reference. Nonetheless, this "kindness" may comes from the weights w_Y we derived for our setup, so it's worth keeping in mind that ν might be overly permissive in some cases.

Another observation: structures that show a bounce-back or are overtaken by SCF-only—such as Ti_2O (Figs. 4, 20, and 21)—display the same behavior in the EOS metrics. Figure 8 illustrates this in the Δ metric, and the trend holds for ν (Fig. 25) and ε (Fig. 26) in Appendix B. This consistency makes sense: if refinement parameters are overtaken by SCF-only, it would be surprising for the EOS-comparison metrics to show the opposite trend. However, even when the refinement plateaus and stops improving, it still reaches the "excellent" threshold in all three metrics (just barely for the Δ metric). Thus, even if the approximation appears to cease functioning beyond that point, the result remains adequate. The only exception is Be_2O, where the metric ν does not reach the "excellent" threshold due to the refinement plateau effect. Another reason one might consider the ν metric too "kind" is that we would expect it to detect this type of anomaly.

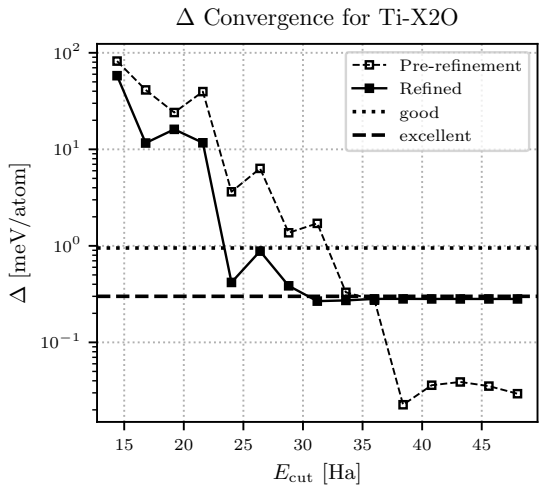

Figure 8: Convergence of the metric Δ, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. After a certain cutoff, the SCF-only method outperforms refinement in precision.

All structures reach the "excellent" threshold at certain cutoffs, except WO_3. Without refinement, WO_3 doesn't even meet the "good" threshold under the Δ metric. However, with refinement, WO_3 not only meets the "good" threshold under the Δ metric but also achieves "excellent." WO_3 fails to reach "excellent" under the Δ metric without refinement (Fig. 9), though it does under ε and ν (Figs. 27, 28 in the appendix B). This shows that even when a structure cannot properly converge by Δ alone, refinement can still boost precision enough to pass the "good" and "excel-

lent" threshold.

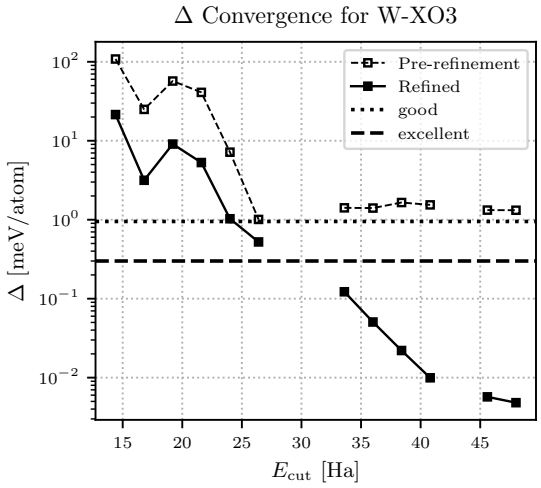

Figure 9: Convergence of the metric Δ, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. The refinement method demonstrates clear convergence advantages across the tested E_{cut} range.

To compare metrics across all structures, Figure 10 presents box plots of each metric versus the cutoff factor $\{E_{\mathrm{cut}}\text{-factor}_i\}_{i=1}^{N}$, showing the median, interquartile range, and outliers. Across all metrics, the refinement protocol converges more quickly than SCF-only.

For the Δ metric, refinement's IQR increases roughly linearly with the cutoff factor, while SCF-only's IQR remains nearly constant. Outliers are also more pronounced under refinement.

The ν metric shows a similar pattern: refinement's IQR grows—in particular, it spikes at the highest cutoff factors—whereas SCF-only stays stable. Again, refinement has more extreme values.

By contrast, the ε metric behaves differently: refinement's IQR initially increases but then falls to small values at high cutoffs, while SCF-only's IQR steadily grows and remains large. Outliers under ε are only slightly more noticeable for refinement and are otherwise comparable between the two methods. At an E_{cut}-factor of 0.275, the refinement protocol's median values for all three metrics lie below the "excellent" threshold. Even at 0.225, its medians reach the "good" threshold. In contrast, the SCF-only method requires an E_{cut}-factor of 0.35 to bring all three medians below "excellent," and 0.3 to drop them under "good." At an E_{cut}-factor of 0.30, the refinement protocol's median values for all three metrics lie below the "excellent" threshold. Even at 0.225, its medians reach the "good" threshold. In contrast, the SCF-only method requires an E_{cut}-factor of 0.35 to bring all three medians below "excellent," and 0.325 to drop them under "good."

We computed the mean of the $E_{\mathrm{cut}}^{\mathrm{high}}$ values across

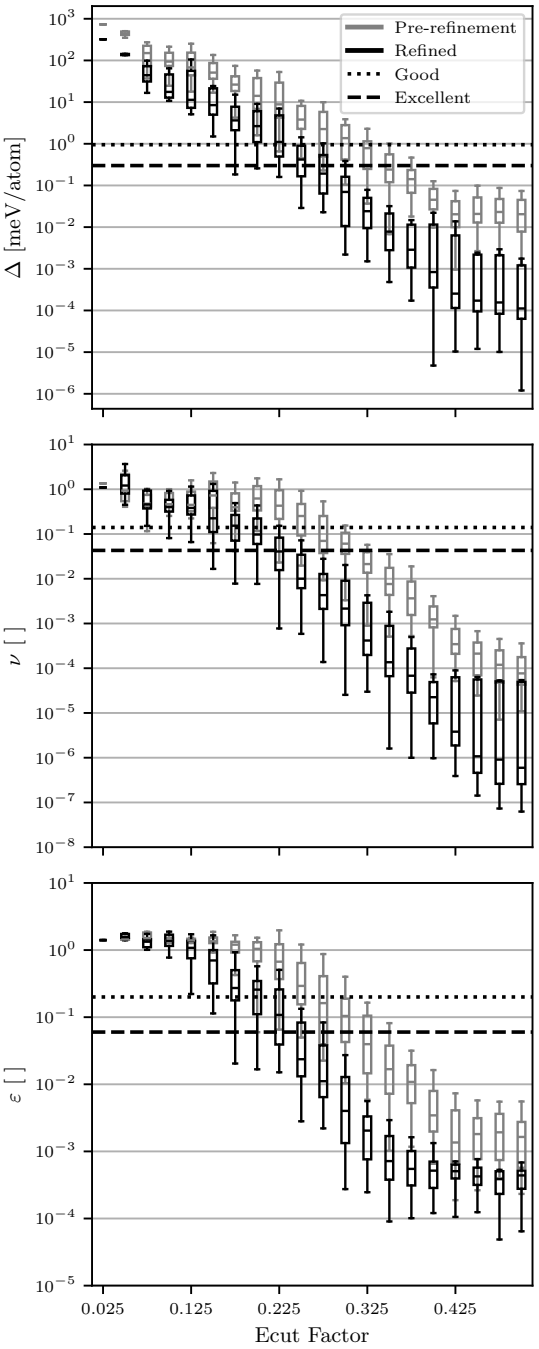

Figure 10: Box plot of Δ, ν and ε deviations versus cutoff factor $\{E_{\mathrm{cut}}\text{-factor}_i\}_{i=1}^{N}$, showing the median, interquartile range, and outliers for all structures to compare metrics convergence on a common relative scale. The refinement method is shown in black; the SCF-only method in gray.

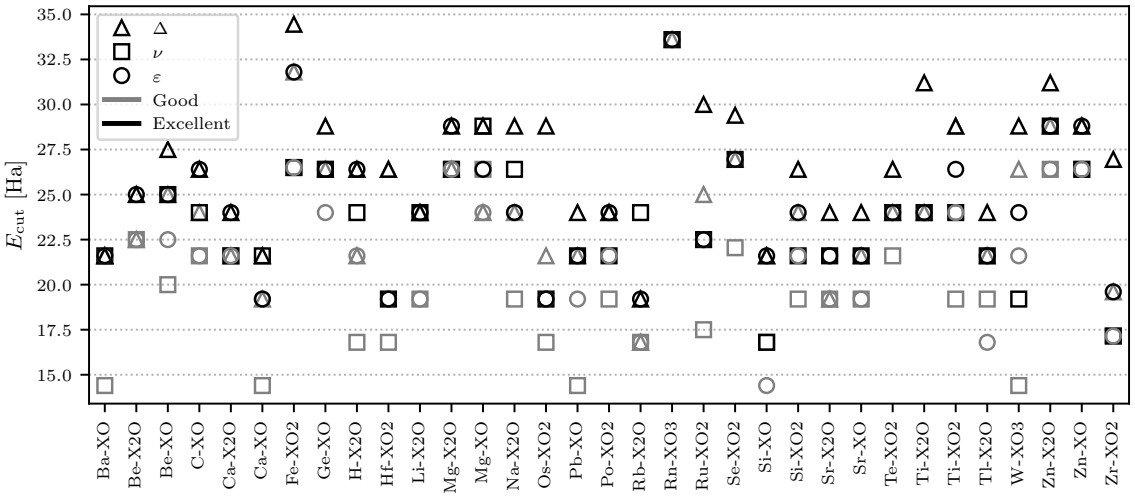

Figure 11: Cutoff energy E_{cut} required to reach the "good" and "excellent" thresholds for the Δ, ε and ν metrics, in function of the benchmark set.

all materials, which is 96.66 Ha. From this rough estimate, achieving "excellent" convergence on all three metrics requires an average E_{cut} of 34 Ha without refinement versus 27 Ha with refinement. To reach "good" convergence across all metrics, an average E_{cut} of 29 Ha is required without refinement versus 22 Ha with refinement.

Figure 11 provides a concise overview of the cutoff energies (E_{cut}) needed to reach "good" and "excellent" thresholds for the Δ, ε, and ν metrics across our benchmark set. For each material, one can directly read off the minimum E_{cut} required to achieve the desired level of precision under each metric.

4.3 Pattern ?

In this section, we look for patterns and groups of materials where the refinement method gives the greatest benefit. To quantify the boost from refinement in the critical region between the "good" and "excellent" thresholds for refinement, we define, for each metric $M \in \{\Delta, \varepsilon, \nu\}$, an improvement score $S(M)$ as the geometric mean of the ratio between SCF-only and refined values at cutoff factors $f \in \{0.225, 0.250, 0.275\}$:

$$S(M) = \sqrt[3]{\prod_{f\in\{0.225,0.250,0.275\}} \frac{M_{\text{no-ref}}(f)}{M_{\text{ref}}(f)}}$$

Here, $M_{\text{no-ref}}(f)$ and $M_{\text{ref}}(f)$ are the metric values without and with refinement at factor f.

The boxplot in Figure 12 displays the $S(\Delta)$) improvement scores for stoichiometries X_2O, XO, XO_2, and XO_3. XO yields the highest mean improvement. In the appendix B the figures 29, 30 shows analogous plots for $S(\epsilon)$, and $S(\nu)$, where no configuration consistently outperforms XO or XO_2. XO_3 is excluded from analysis due to its single sample. Across all three metrics—var, ϵ, and ν—the XO structure achieves slightly higher improvement scores, although its performance closely matches that of XO_2, preventing a definitive conclusion in favor of either. In contrast, X_2O clearly underperforms after refinement, yielding the lowest improvement scores.

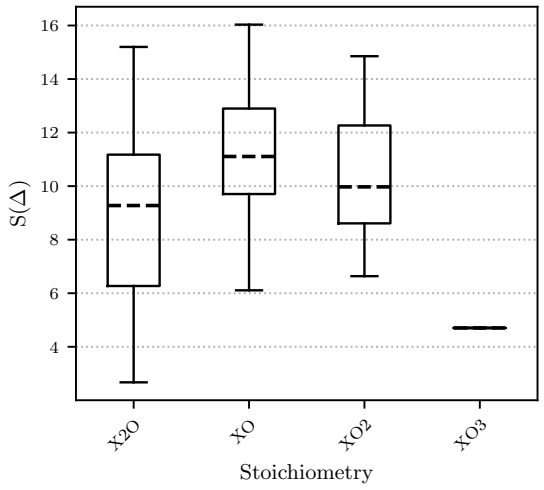

Figure 12: Boxplot of $S(\Delta)$, improvement scores by stoichiometry (X_2O, XO, XO_2, XO_3).

As shown in Figure 13, the scatter of improvement score versus k-point exhibits a positive trend: higher k-point densities correspond to higher improvement scores. In the appendix B figure 31 shows the same behavior for δ, while 32 reveals only a weak, inconclusive trend for ν.

Following the same procedure, we plotted improvement score against electron count, bulk modulus B_0, and equilibrium volume V_0. No clear trends emerged for these parameters; although a slight pattern may ex-

ist for electron count, the data dispersion prevents any definitive conclusion.

Note that the improvement metric was defined informally and may not represent a robust performance measure; accordingly, these results should be treated with caution and validated using more rigorous metrics.

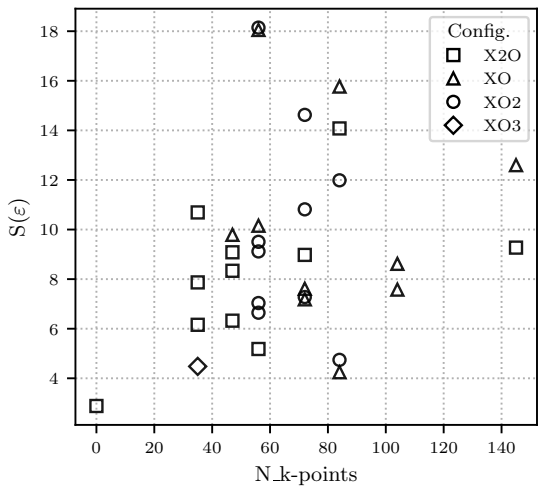

Figure 13: Improvement score vs number of k-points for $S(\varepsilon)$.

4.4 Time Analysis

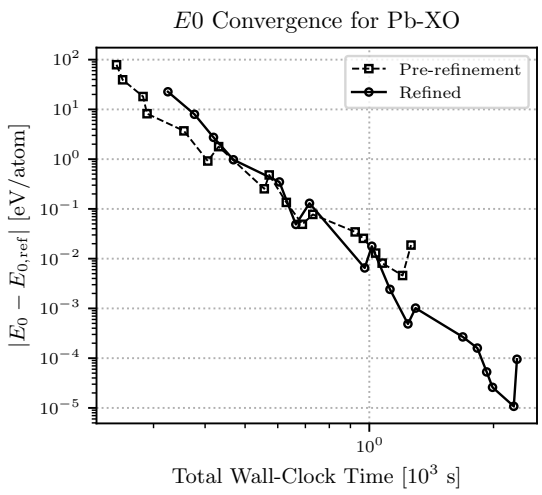

Figure 14: Energy convergence in function of the Wall-Clock time. The energy has been scale with respect to the number of atoms in its cell. The refinement method is shown with a solid line; the SCF-only method with a dashed line.

Although refinement improves data precision, it adds computational cost. This section evaluates whether precision gains justify the extra computation time. We plot energy convergence versus time for pure SCF and for SCF with refinement (i.e., SCF+time to solve 5). Figure 14 shows E_0 convergence over time. Refinement introduces initial overhead but becomes advantageous beyond a certain accuracy. This pattern varies: Figures 33 and 34 show cases where pure SCF reaches the same precision faster. Figure 15 shows a box plot of energy convergence versus time. The two methods perform similarly, with refinement slightly faster on average and gaining an edge at higher precision levels, though fewer SCF runs reach that precision so the top bins are less reliable. We repeated this analysis for metrics Δ, ν, and ε (see Fig. 38,37 and 36 in appendix B) and found no consistent trend: SCF is faster in some precision ranges and refinement in others. The interquartile ranges overlap substantially, making it hard to claim a clear wall-clock time winner. In summary, refinement offers a modest average speed gain but no decisive timing advantage overall.

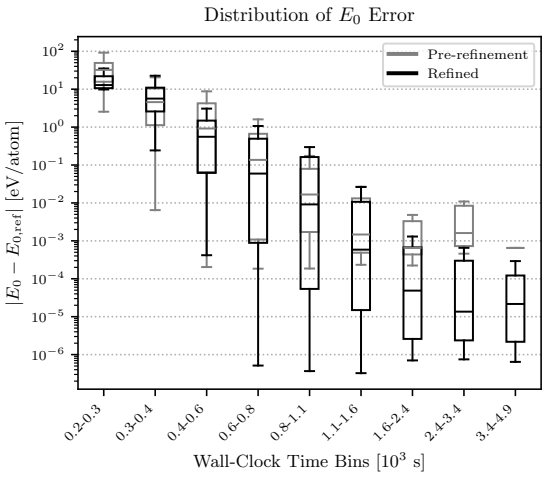

Figure 15: Box plot of energy convergence in function of the Wall-Clock time. The energy has been scale with respect to the number of atoms in its cell. The refinement method is shown with a solid line; the SCF-only method with a dashed line.

5 Conclusion

We benchmarked DFTK's two-stage refinement protocol on thirty-seven insulating oxides and compared it to standard SCF calculations. Refinement consistently speeds up convergence of Birch–Murnaghan EOS parameters and the Δ, ε and ν metrics, reducing the average cutoff for "excellent" agreement from 34 Ha to 27 Ha and for "good" agreement from 29 Ha to 22 Ha. Notably, WO_3 fails the "good" Δ threshold under pure SCF but achieves "excellent" agreement with refinement.

Our time-to-solution analysis reveals only a modest average wall-clock benefit: refinement surpasses SCF

at high precision but does not consistently shorten the total runtime.

Future work should develop stronger metrics to evaluate the refinement protocol's effectiveness and test additional structural parameters—such as lattice geometry—to determine their impact on refinement.

Appendix A ν–Metric Weights

Since the weights and thresholds in [2] aren't tailored to our setup, we derived them using the same methodology (see section 3 and section 8 of [2]). We applied a bootstrap procedure: for each material in our reference set, we created 500 pseudo-datasets by adding Gaussian noise of 1meV/atom to the $E_{\text{cut}}^{\text{high}}$ energies. Fitting a Birch–Murnaghan EOS to each dataset produced distributions of V_0, B_0, and B_0'. Averaging the standard deviations over all materials gave sensitivities σ_{V_0}, σ_{B_0}, and σ_{B_1}. We set $w_{V_0} = 1$ by convention and computed

$$w_{B_0} = \frac{\sigma_{V_0}}{\sigma_{B_0}} \approx 0.0568, \quad w_{B_1} = \frac{\sigma_{V_0}}{\sigma_{B_1}} \approx 0.2755.$$

To define convergence thresholds for our ν metric, we started from the established ε thresholds ($\varepsilon \leq 0.06$ for excellent, $\varepsilon \leq 0.2$ for good) and measured their correlation, finding

$$\nu \approx 0.7\,\varepsilon.$$

This yielded $\nu < 0.043$ for excellent agreement and $\nu < 0.14$ for good agreement. These data-driven weights and thresholds are used in all subsequent analyses.

Appendix B Additional Figures

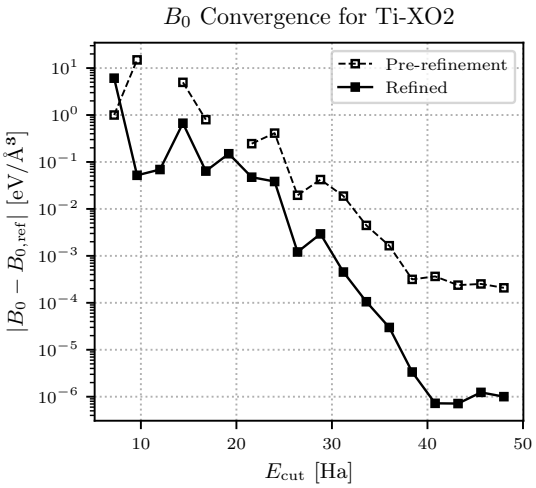

Figure 16: Absolute difference of B_0 from the reference value computed at a high cutoff $E_{\text{cut}}^{\text{high}}$, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. The refinement method demonstrates clear convergence advantages across the tested E_{cut} range.

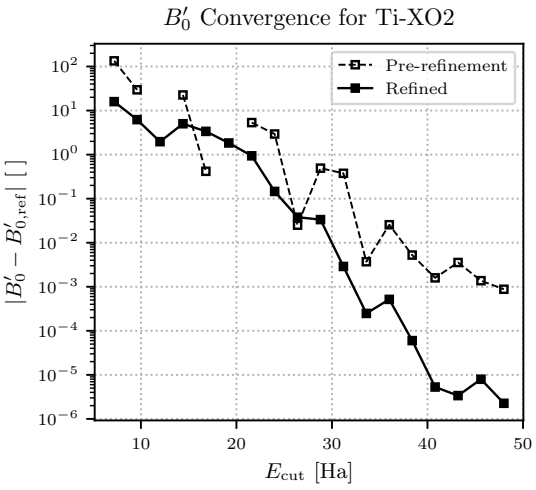

Figure 17: Absolute difference of B_0' from the reference value computed at a high cutoff $E_{\text{cut}}^{\text{high}}$, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. The refinement method demonstrates clear convergence advantages across the tested E_{cut} range.

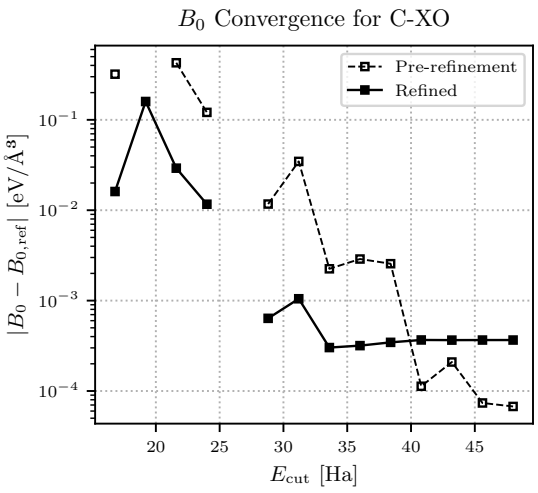

Figure 18: Absolute difference of B_0 from the reference value computed at a high cutoff $E_{\mathrm{cut}}^{\mathrm{high}}$, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. After a certain cutoff, the SCF-only method outperforms refinement in precision.

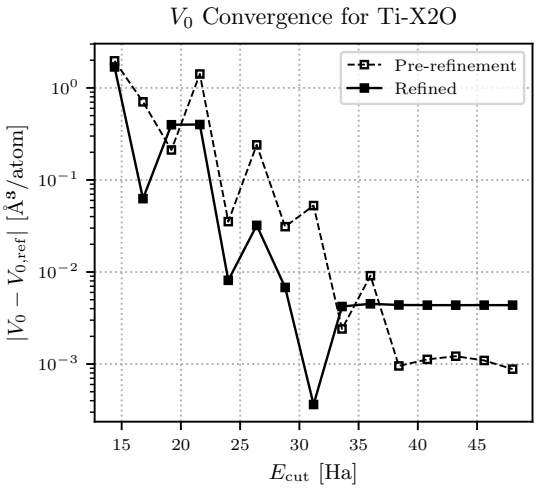

Figure 20: Absolute difference of V_0 from the reference value computed at a high cutoff $E_{\mathrm{cut}}^{\mathrm{high}}$, plotted as a function of E_{cut}. Volumes are scaled with respect to the number of atoms in the simulation cell. The refinement method is shown with a solid line; the SCF-only method with a dashed line. After a certain cutoff, the SCF-only method outperforms refinement in precision.

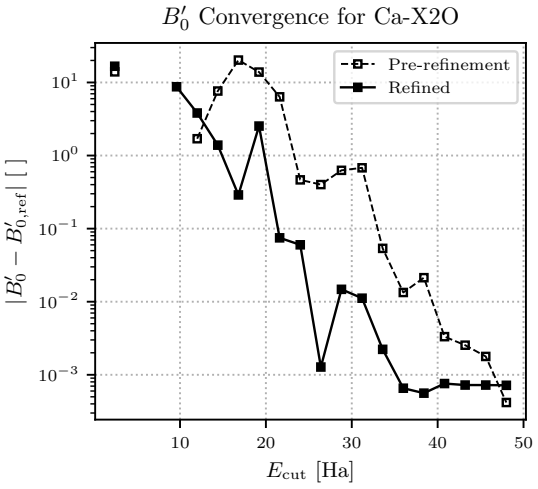

Figure 19: Absolute difference of B_0' from the reference value computed at a high cutoff $E_{\mathrm{cut}}^{\mathrm{high}}$, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. After a certain cutoff, the SCF-only method outperforms refinement in precision.

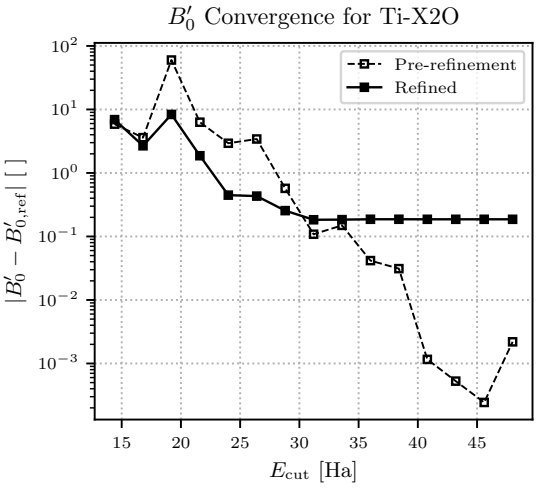

Figure 21: Absolute difference of B_0' from the reference value computed at a high cutoff $E_{\mathrm{cut}}^{\mathrm{high}}$, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. After a certain cutoff, the SCF-only method outperforms refinement in precision.

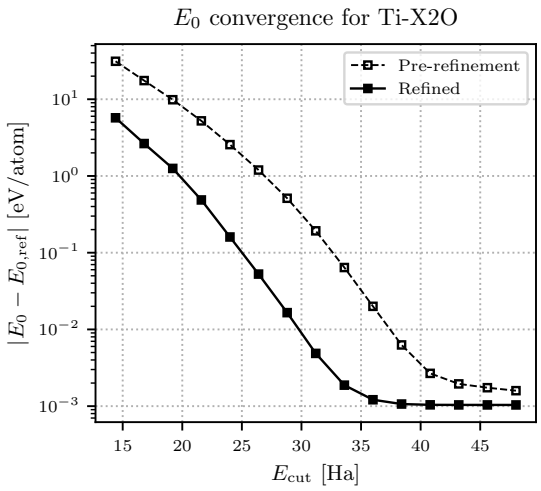

Figure 22: Absolute difference of E_0 from the reference value computed at a high cutoff $E_{\mathrm{cut}}^{\mathrm{high}}$, plotted as a function of E_{cut}. Energies are scaled with respect to the number of atoms in the simulation cell. The refinement method is shown with a solid line; the SCF-only method with a dashed line. The refinement method demonstrates clear convergence advantages across the tested E_{cut} range.

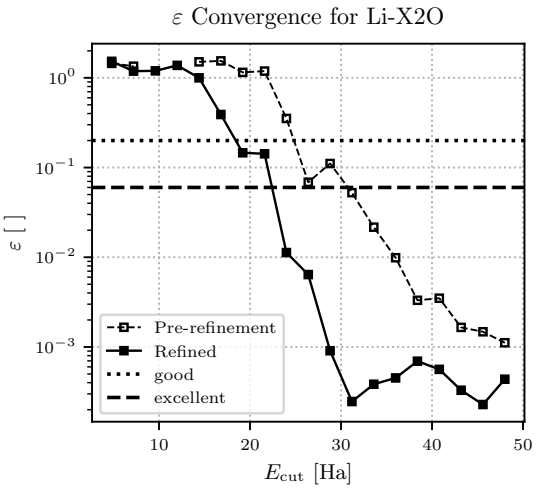

Figure 24: Convergence of the metric ε, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. The refinement method demonstrates clear convergence advantages across the tested E_{cut} range.

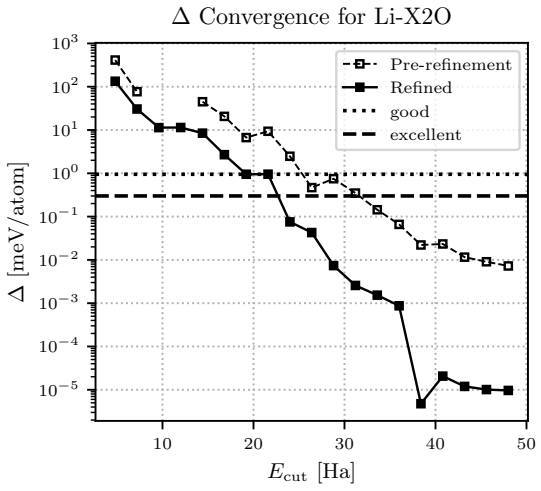

Figure 23: Convergence of the metric Δ, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. The refinement method demonstrates clear convergence advantages across the tested E_{cut} range.

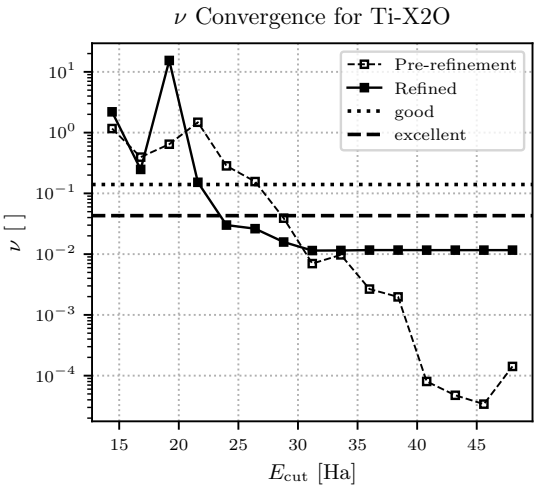

Figure 25: Convergence of the metric ν, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. After a certain cutoff, the SCF-only method outperforms refinement in precision.

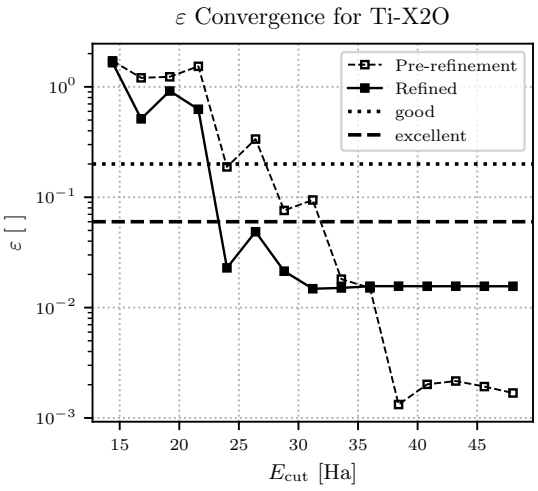

Figure 26: Convergence of the metric ε, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. After a certain cutoff, the SCF-only method outperforms refinement in precision.

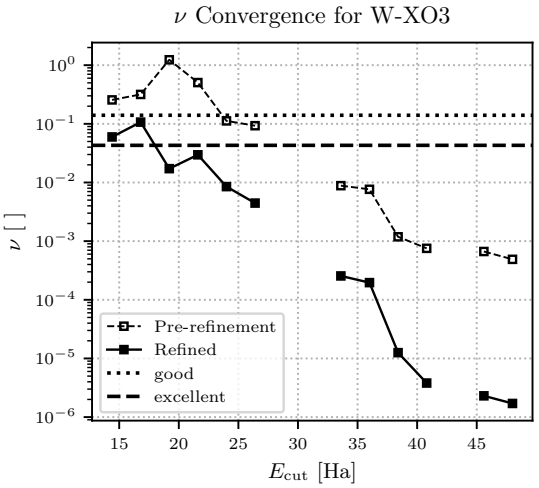

Figure 28: Convergence of the metric ν, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. The refinement method demonstrates clear convergence advantages across the tested E_{cut} range.

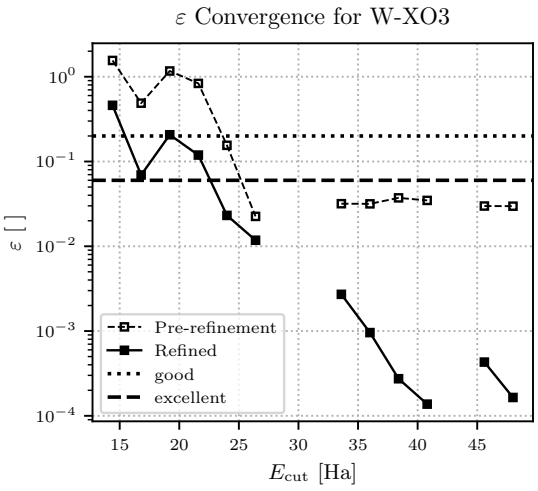

Figure 27: Convergence of the metric ε, plotted as a function of E_{cut}. The refinement method is shown with a solid line; the SCF-only method with a dashed line. The refinement method demonstrates clear convergence advantages across the tested E_{cut} range.

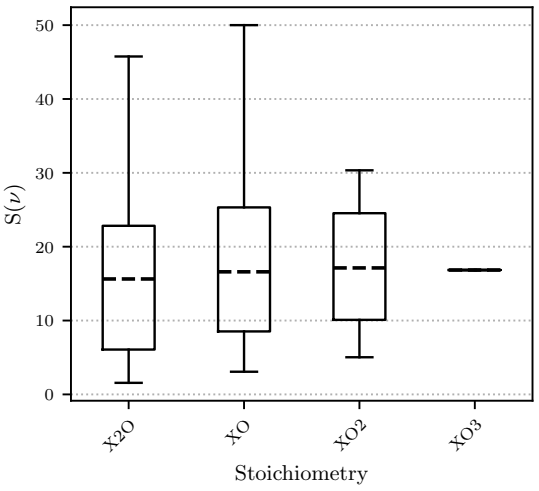

Figure 29: Boxplot of $S(\nu)$, improvement scores by stoichiometry (X_2O, XO, XO_2, XO_3).

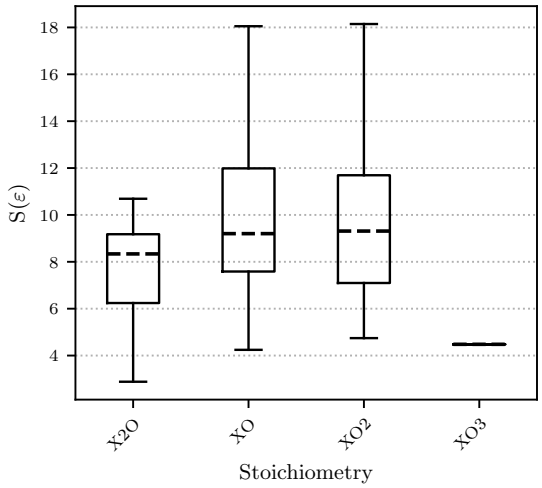

Figure 30: Boxplot of $S(\varepsilon)$, improvement scores by stoichiometry (X_2O, XO, XO_2, XO_3).

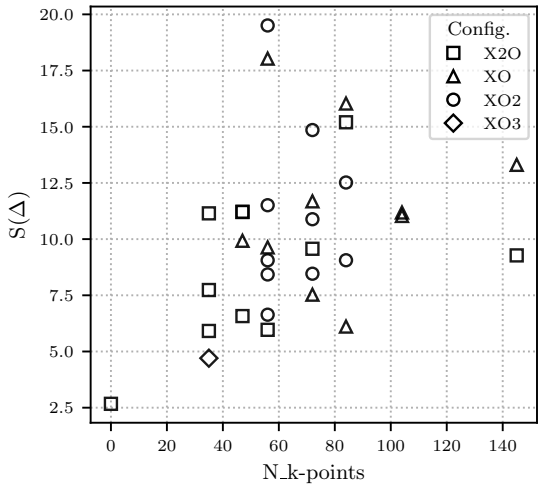

Figure 31: Improvement score vs number of k-points for $S(\Delta)$.

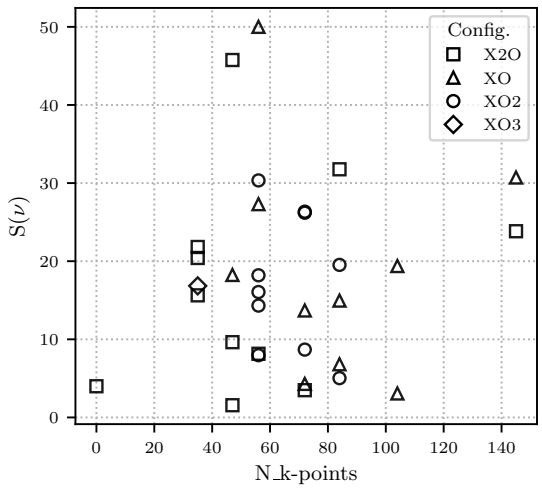

Figure 32: Improvement score vs number of k-points for $S(\nu)$.

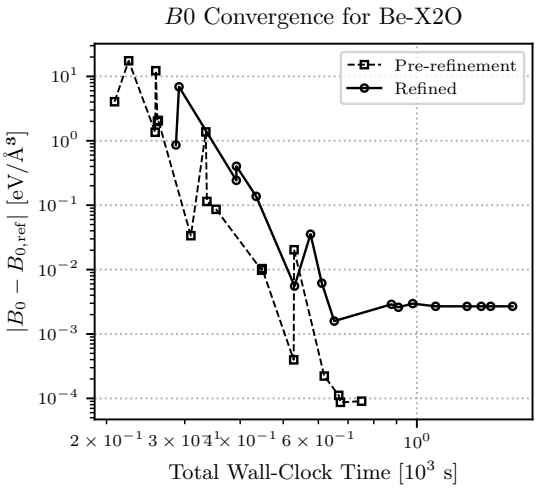

Figure 33: Energy convergence in function of the Wall-Clock time. The energy has been scale with respect to the number of atoms in its cell. The refinement method is shown with a solid line; the SCF-only method with a dashed line.

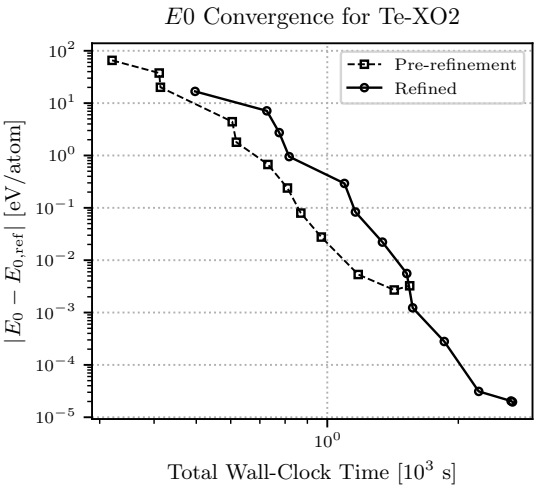

Figure 34: Energy convergence in function of the Wall-Clock time. The energy has been scale with respect to the number of atoms in its cell. The refinement method is shown with a solid line; the SCF-only method with a dashed line.

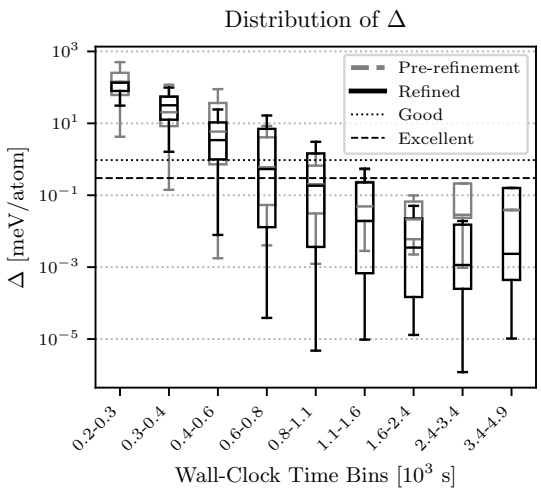

Figure 35: Box plot of Δ metric convergence in function of the Wall-Clock time. The refinement method is shown with a solid line; the SCF-only method with a dashed line.

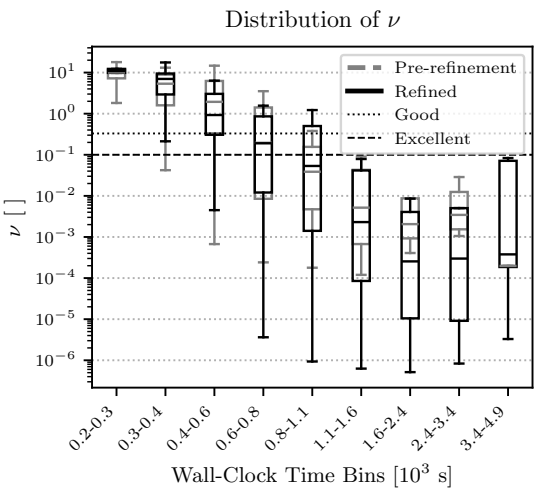

Figure 37: Box plot of ν metric convergence in function of the Wall-Clock time. The refinement method is shown with a solid line; the SCF-only method with a dashed line.

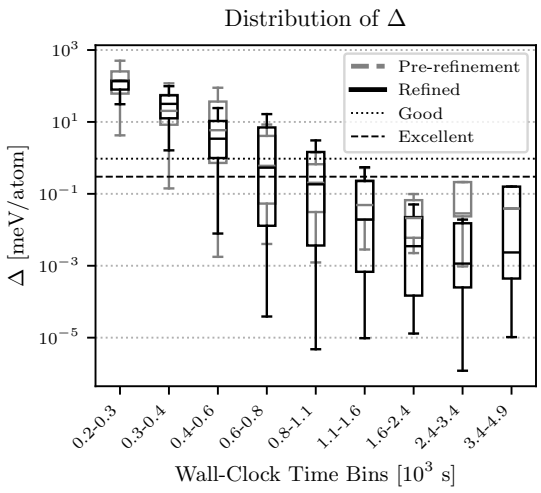

Figure 36: Box plot of Δ metric convergence in function of the Wall-Clock time. The refinement method is shown with a solid line; the SCF-only method with a dashed line.

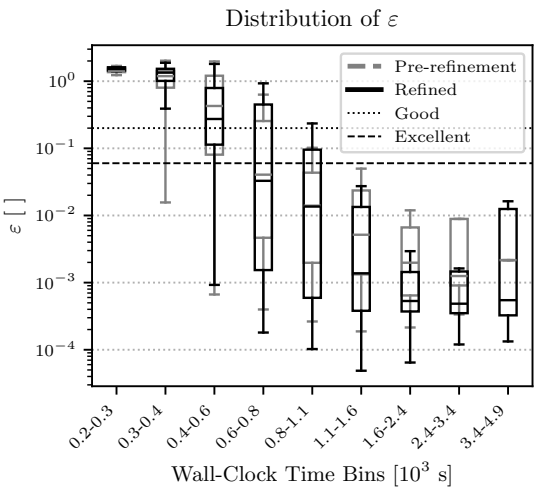

Figure 38: Box plot of ε metric convergence in function of the Wall-Clock time. The refinement method is shown with a solid line; the SCF-only method with a dashed line.

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